

Copula / Probit Transforms and the Autocorrelated Gaussian Prior for Time-Varying Parameters in the Spatial Skew- t Model: Code-to-Text Correspondence

Technical Note — `transform_ar_construction`

Abstract

We document and verify the construction by which the time-varying parameters of the spatial skew- t process acquire their autocorrelation. The governing idea is stated in one sentence:

“For each time-varying parameter, we transform the parameter to obtain a standard normal marginal distribution, place a Gaussian prior with autocorrelation on the transformed parameter, and then transform back to the appropriate marginal distribution for the skew- t process.”

We locate the three operations of this recipe in the `ar2` source tree, give the exact mathematical form of each transform and its inverse CDF, and confirm that the implementation is faithful to the text. The key implementation detail that makes the statement true—namely that the autocorrelated Gaussian prior is normalised to unit marginal variance at *every* time point so the standard-normal marginal is preserved—is made explicit.

1 The three-step recipe and where it lives in the code

There are three time-varying parameters, each indexed by knot k and day $t \in \{1, \dots, n_t\}$:

- the partition precision τ_{kt} (`tau`);
- the skew random effect z_{kt} (`z`);
- the knot location $w_{kt} = (w_{kt}^{(1)}, w_{kt}^{(2)})$ (`knots`).

Each parameter follows the identical three-step recipe. Table 1 maps the sentence onto the source files.

Text concept	Code operation	Location
① transform $\rightarrow$ standard normal marginal	forward copula / probit: <code>gamma.cop</code> , <code>hn.cop</code> , <code>transform\$probit</code>	<code>auxfunctions.R</code> L25–33, L78–81
② Gaussian prior with autocorrelation	stationary AR(2) conditional normal densities, <code>ar2_conditional_log_density</code>	<code>auxfunctions_ar2.R</code> L56–102
③ transform back $\rightarrow$ skew- t marginal	inverse copula / inv. probit: <code>gamma.invcop</code> , <code>hn.invcop</code> , <code>transform\$inv.probit</code>	<code>auxfunctions.R</code> L29–33

Table 1: Correspondence between the textual recipe and the implementation. All paths are relative to `code/R/ar2/`.

The Metropolis updates that drive these parameters live in `update_params_ar2.R`: `updateTauTS_AR2`, `updateZTS_AR2`, and `updateKnotsTS_AR2`. In each, the proposal and acceptance are carried out *on the transformed (Gaussian) scale*, which is natural because the prior is Gaussian there.

2 The transform tools

Throughout, Φ and φ denote the standard normal CDF and PDF, and $\Phi^{-1} = \text{qnorm}$.

Definition 1 (Gaussian copula transform). For a target marginal CDF F , the object `transform$copula` (`auxfunctions.R` L25–28) and its inverse `transform$inv.copula` (L29–32) implement

$$\text{cop}_F(x) = \Phi^{-1}(F(x)), \quad \text{cop}_F^{-1}(u^*) = F^{-1}(\Phi(u^*)). \quad (1)$$

Definition 2 (Probit transform). For $\text{Unif}(l, u)$, `transform$probit` (L14–17) and `transform$inv.probit` (L18–22) implement

$$\text{probit}(w) = \Phi^{-1}\left(\frac{w-l}{u-l}\right), \quad \text{probit}^{-1}(w^*) = (u-l)\Phi(w^*) + l. \quad (2)$$

The probit pair is the copula special case of (1) with $F(w) = (w-l)/(u-l)$, the $\text{Unif}(l, u)$ CDF. The named copulas instantiated in `auxfunctions.R` (L78–81) are

$$\text{gamma.cop} = \text{cop}_{F_{\text{Gamma}}}, \quad \text{gamma.invcop} = \text{cop}_{F_{\text{Gamma}}}^{-1}, \quad \text{hn.cop} = \text{cop}_{F_{\text{HN}}}, \quad \text{hn.invcop} = \text{cop}_{F_{\text{HN}}}^{-1}.$$

Lemma 1 (Standard-normal marginal). *If X has continuous marginal CDF F , then $\text{cop}_F(X) = \Phi^{-1}(F(X)) \sim \mathcal{N}(0, 1)$, since $F(X) \sim \text{Unif}(0, 1)$ and Φ^{-1} maps $\text{Unif}(0, 1)$ to $\mathcal{N}(0, 1)$. Conversely $\text{cop}_F^{-1}(U^*) = F^{-1}(\Phi(U^*))$ has marginal F whenever $U^* \sim \mathcal{N}(0, 1)$.*

Lemma 1 is exactly step ① (forward) and step ③ (backward) of the recipe. It also pins down the requirement that makes the recipe correct: in step ③ the back-transform reproduces the intended marginal F *only if* the transformed variate is $\mathcal{N}(0, 1)$. Section 4 shows the AR(2) prior is engineered to guarantee this at every t .

3 Per-parameter marginals, forward transforms, and inverse CDFs

3.1 Precision τ : Gamma marginal

The marginal is $\tau \sim \text{Gamma}(\alpha, \beta)$ with shape $\alpha = \text{tau.alpha}$ and rate $\beta = \text{tau.beta}$.

Forward (to $\mathcal{N}(0, 1)$). `update_params_ar2.R` L19, `tau.star <- gamma.cop(tau, tau.alpha, tau.beta)`:

$$\tau^* = \Phi^{-1}(F_{\text{Gamma}}(\tau; \alpha, \beta)), \quad F_{\text{Gamma}}(\tau; \alpha, \beta) = \frac{\gamma(\alpha, \beta\tau)}{\Gamma(\alpha)}, \quad (3)$$

where $\gamma(\cdot, \cdot)$ is the lower incomplete gamma function.

Inverse CDF (back to $\mathbb{R}_+$). `update_params_ar2.R` L31, `can.tau <- gamma.invcop(can.tau.star, tau.alpha, tau.beta)`:

$$\tau = F_{\text{Gamma}}^{-1}(\Phi(\tau^*); \alpha, \beta) = Q_{\text{Gamma}}(\Phi(\tau^*)), \quad (4)$$

with $Q_{\text{Gamma}} = \text{qgamma}$ the gamma quantile function. A floor of 10^{-6} is applied for numerical stability (L32–34).

3.2 Skew effect z : Half-Normal marginal

Conditional on τ , the model takes $z | \tau \sim \text{HN}(0, \sigma)$ with scale $\sigma = 1/\sqrt{\tau}$ (`update_params_ar2.R` L343–344). The half-normal CDF and quantile function are (`auxfunctions.R` L56–73)

$$F_{\text{HN}}(z; \sigma) = 2\Phi(z/\sigma) - 1, \quad F_{\text{HN}}^{-1}(p; \sigma) = \sigma \Phi^{-1}\left(\frac{1+p}{2}\right), \quad z \geq 0. \quad (5)$$

Forward (to $\mathcal{N}(0, 1)$). `z.star <- hn.cop(z, sig):`

$$z^* = \Phi^{-1}(2\Phi(z\sqrt{\tau}) - 1). \quad (6)$$

Inverse CDF (back to $\mathbb{R}_+$). `update_params_ar2.R` L372, `can.z <- hn.invcop(can.z.star, sig):`

$$z = \frac{1}{\sqrt{\tau}} \Phi^{-1}\left(\frac{1+\Phi(z^*)}{2}\right). \quad (7)$$

Remark 1 (A latent inconsistency in `dhn`, off the transform path). The density helper `dhn` returns `dnorm(x, 0, sig)/2` (`auxfunctions.R` L47), i.e. $\varphi(x/\sigma)/(2\sigma)$, whereas the true half-normal density consistent with F_{HN} above is $2\varphi(x/\sigma)/\sigma$ —a factor of four apart. The copula transforms call only `phn` and `qhn`, so `dhn` never enters the AR construction; the discrepancy is a latent bug with no effect on the pipeline.

3.3 Knot location w : Uniform marginal

Each coordinate $j \in \{1, 2\}$ has marginal $w^{(j)} \sim \text{Unif}(l_j, u_j)$ with $l_j = \min.s[j]$, $u_j = \max.s[j]$.

Forward (to $\mathcal{N}(0, 1)$). `update_params_ar2.R` L477–484, `transform$probit:`

$$w^{(j)*} = \Phi^{-1}\left(\frac{w^{(j)} - l_j}{u_j - l_j}\right). \quad (8)$$

Inverse CDF (back to the spatial domain). `update_params_ar2.R` L520–527, `transform$inv.probit:`

$$w^{(j)} = (u_j - l_j) \Phi(w^{(j)*}) + l_j. \quad (9)$$

4 Step ②: the autocorrelated Gaussian prior

Write X_t^* for the transformed series, i.e. any of τ_{kt}^* , z_{kt}^* , $w_{kt}^{(j)*}$. The prior is a stationary, mean-zero, *unit-variance* AR(2) (`auxfunctions_ar2.R` L9–54):

$$X_t^* = \phi_1 X_{t-1}^* + \phi_2 X_{t-2}^* + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2). \quad (10)$$

The decisive design choice is the normalisation $\gamma_0 \equiv 1$ (`auxfunctions_ar2.R` L31). The remaining stationary autocovariances and the innovation variance then follow from the Yule–Walker equations:

$$\gamma_0 = 1, \quad \gamma_1 = \frac{\phi_1}{1 - \phi_2}, \quad \gamma_2 = \phi_1 \gamma_1 + \phi_2, \quad \sigma_\varepsilon^2 = \gamma_0 - \phi_1 \gamma_1 - \phi_2 \gamma_2. \quad (11)$$

Used as a prior, the joint density factorises into the conditional normals (`auxfunctions_ar2.R` L62–92):

$$X_1^* \sim \mathcal{N}(0, 1), \quad (12)$$

$$X_2^* | X_1^* \sim \mathcal{N}(\gamma_1 X_1^*, 1 - \gamma_1^2), \quad (13)$$

$$X_t^* | X_{t-1}^*, X_{t-2}^* \sim \mathcal{N}(\phi_1 X_{t-1}^* + \phi_2 X_{t-2}^*, \sigma_\varepsilon^2), \quad t \geq 3. \quad (14)$$

Stationarity is enforced by the triangle (`check_ar2_stability`, L2–7):

$$|\phi_2| < 1, \quad \phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1. \quad (15)$$

Proposition 1 (Marginal invariance). *Under (10)–(14), every X_t^* has marginal $\mathcal{N}(0, 1)$. Consequently, by Lemma 1, the back-transform of step ③ returns each parameter to its intended marginal (Gamma, HN, or Unif) at every time point.*

Sketch. Equation (12) gives $\text{Var}(X_1^*) = \gamma_0 = 1$. For (13), $\text{Var}(X_2^*) = \gamma_1^2 \text{Var}(X_1^*) + (1 - \gamma_1^2) = 1$. For $t \geq 3$, the Yule–Walker choice (11) of σ_ε^2 is precisely the value that propagates $\text{Var}(X_t^*) = 1$ forward. Hence the marginal variance is 1 for all t ; with zero mean this is $\mathcal{N}(0, 1)$. $\square$

5 Verdict: the code is faithful to the text

The implementation realises each clause of the sentence, and additionally handles two requirements that the prose leaves implicit but that correctness demands.

- (i) **“standard normal marginal” must hold at every t .** The back-transform $F^{-1}(\Phi(X_t^*))$ recovers the correct F only when $X_t^* \sim \mathcal{N}(0, 1)$. The $\gamma_0 = 1$ normalisation (Proposition 1) secures this. In particular, the $t = 2$ conditional (13) uses the stationary bivariate form $\mathcal{N}(\gamma_1 X_1^*, 1 - \gamma_1^2)$ rather than the backward-extrapolated $\mathcal{N}(\phi_1 X_1^*, \sigma_\varepsilon^2)$; the two coincide only when $\phi_2 = 0$, and only the former keeps $\text{Var} = 1$. This is the “Fix 1” branch in `get_ar2_conditional_params` (`auxfunctions_ar2.R` L70–85).
- (ii) **The Metropolis ratio must include the $t+1$ and $t+2$ terms.** Because X_t^* appears as the lag-1 term of the $t+1$ conditional and the lag-2 term of the $t+2$ conditional, a proposal for X_t^* changes three factors of the joint prior. The updates add all three contributions to the acceptance ratio (“Fix 2”, e.g. `update_params_ar2.R` L62–95); omitting the lag-2 term would break detailed balance and bias the lag-2 autocorrelation away from γ_2 .

Finally, the prior-predictive simulators are duals of the same maps: `makeTauAR2`, `makeZAR2`, and `makeKnotsAR2` (`auxfunctions_ar2.R` L153–193) draw a unit-variance stationary AR(2) path X^* via `simulate_ar2_standard` and then apply the corresponding inverse copula / inv. probit. The `*ARP` variants generalise the AR(2) to a general AR(p), solving the discrete Lyapunov equation for the stationary state covariance while preserving $\gamma_0 = 1$; the logic is identical.

Conclusion. The code implements “transform $\rightarrow$ standard normal $\rightarrow$ autocorrelated Gaussian prior $\rightarrow$ back to the skew- t marginal” exactly as written, with the three parameters (τ, z, w) mapped through Gamma, Half-Normal, and Uniform copula marginals respectively. The only transform-irrelevant defect is the unused `dhn` normalising constant.